

Merchant Revenue Forecasting

Choosing a governable default for 12-month merchant-level planning

Self-directed portfolio project using synthetic data

BUSINESS PROBLEM
Which forecasting method should Finance use as the default 12-month merchant-revenue forecast, and when should merchant-level performance justify an exception?

PORTFOLIO 50 synthetic merchants	HISTORY 60 months (2020–2024)	OBSERVATIONS 3,000 merchant-months	PRIMARY METRIC Unweighted mean MAPE
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Finance needs a repeatable forecast across merchants with different growth, seasonality, and volatility. The 2024 result uses 48 months of training history and a common 12-month holdout. Rolling-origin checks forecast 2022, 2023, and 2024 from each prior year-end. Seasonal Naïve is the planning baseline. Driver Scenario Regression uses origin-safe planning assumptions, not realized holdout drivers. MAPE is an unweighted mean of merchant-level error, not a revenue-weighted portfolio score.

2024 HOLDOUT RESULT
Holt achieved the lowest mean merchant-level MAPE at **8.64%**, compared with **12.47%** for Seasonal Naïve. That is a 3.83 percentage-point improvement, or a 30.7% relative reduction in forecast error.

30.7%
relative reduction versus Seasonal Naïve on the 2024 holdout

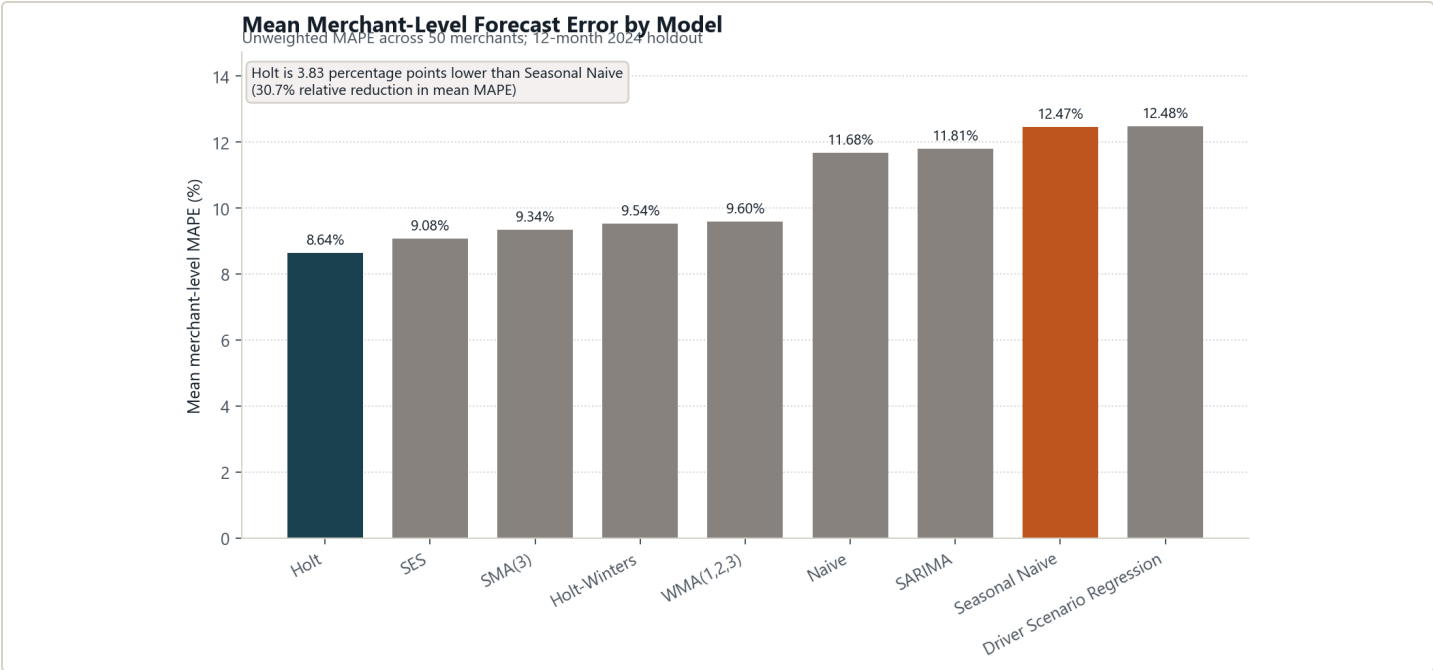


Figure 1. Unweighted mean of merchant-level MAPE across 50 merchants, 12-month 2024 holdout. Holt is highlighted as the 2024 champion; Seasonal Naïve is the planning baseline.

Interpretation: Additional complexity did not automatically improve accuracy. After leakage was removed, Driver Scenario Regression had a 12.48% mean MAPE and SARIMA 11.81%, while simpler smoothing methods occupied the strongest 2024 positions. Holt was best for 17 of 50 merchants (34%). Rolling-origin results do not support treating Holt as a time-robust default: SES had the lowest average error across 2022–2024.

Merchant Variation, Recommendation, and Governance

Use the 2024 champion with challengers, then manage exceptions by merchant

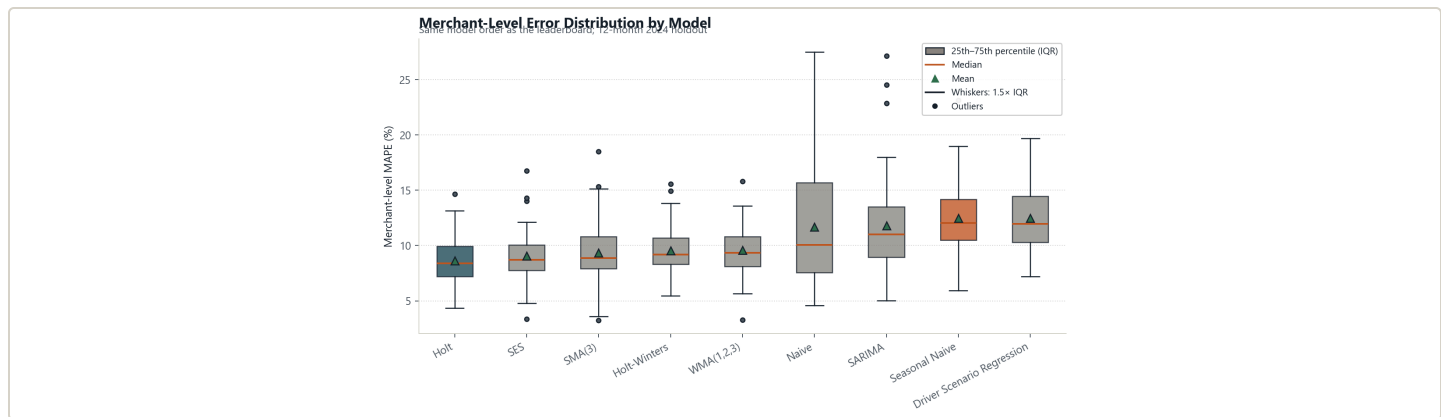


Figure 2. Merchant-level MAPE distribution, same order as Figure 1. Whiskers are $1.5 \times$ IQR. Holt combined low average error with a comparatively contained distribution versus weaker methods, although no method eliminated variation across merchants.

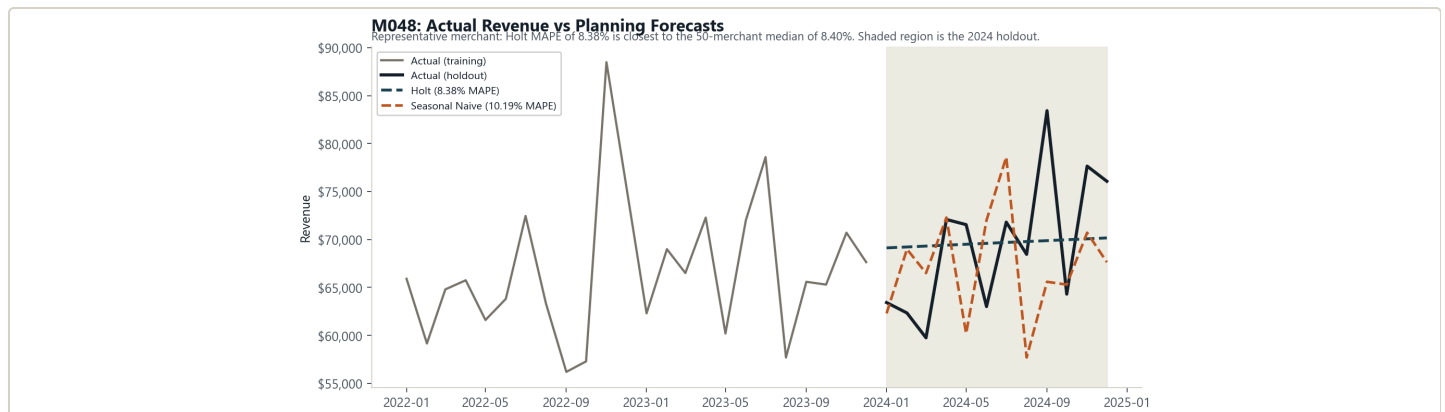


Figure 3. Merchant M048 is shown because its Holt MAPE (8.38%) is closest to the 50-merchant median (8.40%). Only the 2024 champion and the Seasonal Naive baseline are plotted.

Planning Recommendation

- Treat Holt as the 2024 champion, not a permanent default. SES had the lowest average error across the 2022–2024 rolling windows (9.75% vs 9.88% for Holt), because Holt deteriorated in the 2022 origin (12.31%).
- Retain Seasonal Naive as the minimum-performance benchmark for every planning cycle.
- Run Holt, SES, and SMA(3) as a champion/challenger set on the same cutoff and 12-month horizon.
- Permit a merchant-level override only when another method produces consistently lower error across multiple forecast windows.
- Maintain a high-uncertainty watchlist for merchants with persistently elevated or volatile error; give those names wider ranges, partner input, and explicit upside/downside scenarios.

Limitations

- The dataset is synthetic and does not represent an employer or customer portfolio.
- Results may not generalize to real merchant businesses.
- Primary MAPE is an unweighted mean across merchants, not portfolio-dollar forecast error.
- This work does not claim production deployment or realized business savings.
- Driver forecasts depend on planning assumptions available at the origin.
- Rankings can change as new actuals arrive; confirm selection through rolling-origin backtesting.
- SARIMA and Driver Scenario Regression were not eligible for the 2022 window because they require 36 months of training history.

Takeaway for finance leaders: The best forecasting process is not necessarily the most technically complex. A transparent champion, an objective baseline, rolling-origin checks, and disciplined exception monitoring can provide a more governable planning process than selecting models by sophistication alone.